


# Trigonometry
# Squaring and adding




# MATHS HOTSHOT | SSC CGL 2026

by PARMAR SSC

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## Squaring and Adding ($\sin^2\theta + \cos^2\theta = 1$)

$\frac{a}{b} \Longleftrightarrow \frac{a+b}{a-b}$

### Algebraic Form (Origin)

$$\underset{①}{(ax+by)^2} + \underset{②}{(ay-bx)^2} = \underset{③}{(a^2+b^2)}\underset{④}{(x^2+y^2)}$$

इन चार भागों में से तीन की values दी होती हैं और चौथे की value पूछी जाती है।

★ Used when equal and opposite $2ab$ terms cancel on addition.

$ax+by=5$

$bx-ay=3$

$(a^2+b^2)(x^2+y^2) = 25+9 = 7$

If $a\sin\theta \oplus b\cos\theta = p$
and $a\cos\theta \ominus b\sin\theta = q$

$$a^2+b^2 = p^2+q^2$$

$3\sin\theta+2\cos\theta=p$

$2\sin\theta-3\cos\theta=q$

$13=(9+4)=p^2+q^2$

$(a^2+b^2)(5^2+2^2) = p^2+q^2$

$x^2+y^2 = 4+16$

1. The elimination of $\theta$ from $x\cos\theta - y\sin\theta = 2$ and $x\sin\theta + y\cos\theta = 4$ will give

समीकरण $x\cos\theta - y\sin\theta = 2$ and $x\sin\theta + y\cos\theta = 4$ में से अगर $\theta$ को ख़त्म करने पर बचेगा-

a) $x^2 + y^2 = 20$
b) $3x^2 + y^2 = 20$
c) $x^2 - y^2 = 20$
d) $3x^2 - y^2 = 10$

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$x^2+y^2= a^2+b^2$

2. If $x = a\sin\theta - b\cos\theta$, $y = a\cos\theta + b\sin\theta$, then which of the following is true?

अगर $x = a\sin\theta - b\cos\theta, y = a\cos\theta + b\sin\theta$ है तो निम्न में से कौनसा सही है?

a) $\frac{x^2}{y^2} + \frac{a^2}{b^2} = 1$ b) $x^2 + y^2 = a^2 - b^2$

c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ d) $x^2 + y^2 = a^2 + b^2$

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3. If $2\sin\theta + 3\cos\theta = 2$, then find the value of $3\sin\theta - 2\cos\theta$.

- **Exam:** SSC CGL Tier-1
- **Date:** 05 December 2022 (Shift 4)

$= t$

$4 + 9 = 4 + t^2$

$3 = t$

4. If $a\sin\theta + b\cos\theta = c$, then what is the value of $a\cos\theta - b\sin\theta$?

- **Exam:** SSC CHSL Tier-1
- **Date:** 11 August 2021 (Shift 3)

$= t$

$a^2 + b^2 = c^2 + t^2$

$\sqrt{a^2 + b^2 - c^2} = t$

5. If $x = 6\cos A + 8\sin A$ and $y = 6\sin A - 8\cos A$, find the value of $x^2 + y^2$.

यदि $x = 6\cos A + 8\sin A$ और $y = 6\sin A - 8\cos A$ है, तो $x^2 + y^2$ का मान ज्ञात करें।

a) 120
b) 150
c) 170
d) 100

$(6^2+8^2)(\sin^2+\cos^2)$

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$\sin\theta = \frac{5}{17}$

$\sin^2\theta + \cos^2\theta = ?$

6. If $sinA = \frac{2}{3}$, find the value of $(3sinA - 4cosA)^2 + (4sinA + 3cosA)^2$.

यदि $sinA = \frac{2}{3}$, तो $(3sinA - 4cosA)^2 + (4sinA + 3cosA)^2$ का मान ज्ञात कीजिए।

a) 25
b) 18
c) 13
d) 34

$3^2 + 4^2 = 5^2 = 25$

$3x(7^2+24^2)$

$= 3x625$

7 If $x = 7\sin A + 24\cos A$ and $y = 7\cos A - 24\sin A$, then what is the value of $3(x^2 + y^2)$?

यदि $x = 7\sin A + 24\cos A$ तथा $y = 7\cos A - 24\sin A$ हो, तो $3(x^2 + y^2)$ का मान क्या है?

a) 1760
b) 2120
c) 1875
d) 1290

# Squaring and Subtracting ($\sec^2\theta - \tan^2\theta = 1 = \operatorname{cosec}^2\theta - \cot^2\theta$)

If $a\sec\theta - b\tan\theta = p$

and $a\tan\theta - b\sec\theta = q$

$\Rightarrow \quad a^2 - b^2 = p^2 - q^2$

$\sin^2+\cos^2=1$

$\sec^2-\tan^2=1$

$\operatorname{cosec}^2-\cot^2=1$

★ Both squared expressions contain the same cross term, $-2ab\sec\theta\tan\theta$. Therefore, it cancels when the second square is subtracted from the first.

$a^2-b^2 = p^2-q^2$

$b^2-a^2 = q^2-p^2$

8. If $a\sec\theta - b\tan\theta = 8$ and $a\tan\theta - b\sec\theta = 6$, and $a^2 - b^2 = 2k$, what is the value of $k$?

a) 14
b) 28
c) 7
d) 50

$a^2 - b^2 = 64 - 36 = 28$

$\Rightarrow 28$
$14$

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9. If $x \csc\theta - y \cot\theta = 6$ and $x \cot\theta - y \csc\theta = 4$, then what is the value of $x^2 - y^2$?

a) 52

b) 10

c) 20

d) 24

$x^2 - y^2 = 6^2 - 4^2 = 36 - 16 = 20$

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10. If $5\sec\theta - 4\tan\theta = \sqrt{13}$ and $5\tan\theta - 4\sec\theta = q$ (where $q > 0$), find the value of $q$.

a) 3
b) 4
c) 2
d) 5

$25 - 16 = 13 - q^2$

$9 = 13 - q^2$

$q^2 = 4$

$q = 2$

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